Waveform Design and Processing in OFDM based Radar-Communication Integrated Systems

Xuanxuan Tian*, Tingting Zhang*, Qinyu Zhang* and Zhaohui Song†
*Communication Engineering Research Center, Shenzhen Graduate School,
Harbin Institute of Technology, Shenzhen, China
Email: tianxuanxuan2008@163.com, {zhangtt, zqy}@hit.edu.cn
†School of Information Science and Technology, East China Normal University, Shanghai, China
Email: zhsong@ce.ecnu.edu.cn

***Abstract*—This paper presents an integrated radar and communication system using a unique platform to meet the requirements of future intelligent transportation networks. A phase-coded orthogonal frequency division multiplex (PC-OFDM) waveform in which the phase sequence being controlled via transmitted messages is given. A radar data processing algorithm that is able to achieve the joint estimation of the range and velocity simultaneously, for the single-scatter-point target is presented within one single transmitted pulse. Moreover, the performance analysis of the radar algorithm is analyzed in detail. Theoretical analysis and numerical results show that, the presented methods can achieve higher-resolution range-velocity profiles of the targets along with the relative high data rate transmissions.**

***Index Terms*—Radar-communication integration, OFDM, phase-coded, high resolution range-velocity profile**

I. Introduction

Due to the trends of similarity of wireless radar and communication hardware platforms, especially on the front ends, "intelligent" waveform design and processing methods that can fulfill the radar and communications simultaneously (known as "RadCom" in literatures) attract the broad attentions [1], [2]. The integrated radar-communication systems using orthogonal frequency division multiplex (OFDM) signals based on the operation at $24\mathrm{GHz}$ have been presented in [2], [3]. The systems using OFDM waveforms for simultaneously performing radar and communication functions on a single hardware platform in order to improve the spectrum efficiency and cost effectiveness. A typical application is the intelligent transportation networks which require the ability of inter-vehicle communications as well as reliable environment sensing, both are critical for intelligent, or even autonomous driving.

OFDM waveforms, along with a coding scheme, are called multicarrier complementary phase-coded (MCPC) signals [4]. A ultra-wideband (UWB) digital system to measure the radar cross section of targets was proposed in [5], where the transmitted waveform was called phase-coded OFDM (PC-OFDM) signal. Due to their advantages on high spectral efficiency, thumbtack-liked ambiguity function [4], good Doppler tolerance [6], flexible waveform structure and easy implementations [7], the PC-OFDM signals are attractive to both academic and industrial researchers.

Various researches have been investigated on range and velocity measurement in the OFDM integrated systems. A classical correlation processing approach was presented in [8], where one subcarrier carried 1 bit information, resulting in a low data rate and high range sidelobes, and concerning less on velocity measurement. A modulation symbol-based processing approach based on the traditional OFDM signals was presented in [2], where the performance analysis of the approach was not discussed in detail. An adaptive pulse compression approach to improve range performance was presented in [9], it introduced a very high computational effort on the cyclic iterative algorithm. A subspace-based algorithm using few OFDM symbols was developed in [10], however, the velocity measurement depended on the range information. A non-linear least squares approach using weighted OFDM modulation scheme was presented in [11], the range performance was improved, but not improve the velocity performance. Two-step Doppler estimation scheme of OFDM radar was proposed in [12], however, oversampling in the frequency domain increased the separation difficulty of subcarriers. A subspace projection method based on the compensated communication information was appeared in [13], however, the design of cyclic prefix was important to the system performance. OFDM linear frequency modulation integrated signal based on fractional Fourier fransform (FRFT) [14] and a random stepped frequency OFDM integrated signal [15] were presented to improve data rate and range resolution, but they were prone to range-Doppler coupling.

In pulsed radar systems, the duration of the burst of pulses that are integrated coherently determines the Doppler resolution [16]. However, the pulse repetition frequency (PRF) introduces the range ambiguity and Doppler ambiguity, and high Doppler sensitivity.

In view of the above, a single pulse combining the OFDM modulation with phase coding technique, uses the transmitted information to control the cyclic-shift value of a phase sequence, resulting in a high data rate. The radar processing algorithm for the continuous OFDM waveforms, as developed in [2], is here put into a formal proof. Moreover this paper gives a more detailed analysis of the algorithm performance for the discrete signal model, to achieve a high resolution range-velocity profile with low computational complexity.

The rest of this paper is organized as follows. The signal model of PC-OFDM integrated waveform is given in II. The radar processing algorithm is explained in III. The perfor-

Authorized licensed use limited to: N.C. State University Libraries - Acquisitions & Discovery S. Downloaded on February 18,2026 at 03:32:02 UTC from IEEE Xplore. Restrictions apply.

978-1-5386-4502-4/17/$31.00 ©2017 IEEE

mance analysis of the algorithm is given in IV. The simulation results are presented in section V. Conclusion of the paper is given in section VI.

II. Signal Model

A. Transmitted Signals

We consider a monostatic radar transmits one single pulse, which is composed of N subcarriers. The transmitted messages on each subcarrier are mapping onto a sequence of M bits. Then the transmitted signal for a single pulse can be written as

$$x(t) = \sum_{n=0}^{N-1}\sum_{m=0}^{M-1} \omega_n a_{nm} \text{rect}(t - mt_b) e^{j2\pi n\Delta f t} e^{j2\pi f_c t} \quad (1)$$

where f_c is the radio frequency (RF) carrier frequency, N and M are the number of subcarriers and codes, respectively, ω_n is the weight of the nth subcarrier, t_b is the code duration, $\Delta f = 1/t_b$ is the subcarrier separation, $B = N\Delta f$ is the signal bandwidth, rect(t) is given by

$$\text{rect}(t) = \begin{cases} 1 & 0 \le t \le t_b \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

and $\{a_{nm}\}_{m=0}^{M-1}$ is the transmitted PC sequence on the nth subcarrier. The construction of PC sequences is through phase coding technique, where the information data controls the cyclic-shift value of a phase sequence.

Specific construction steps are summarized as follows: (i) select an initial phase sequence $\mathbf{s}$ of length M with a better periodic autocorrelation function (ACF). (ii) The transmitted binary data bits $(0, +1)$ of size $(N \times K)$ is generated randomly, and K is assumed to fulfill $2^K \le M$. The K bits information carried on the nth subcarrier converts into the decimal form d_n, which represents the time shift on the nth subcarrier. The time shifts for all subcarriers is represented as $\mathbf{d} = \{d_n, n = 0, \ldots, N-1\}$. (iii) Cyclic shift. $\{a_{nm}\}_{m=0}^{M-1} = \mathbf{s}_{\to d_n}$, where $\mathbf{s}_{\to d_n}$ represents cyclically shifted $\mathbf{s}$ by d_n. Hence, we obtain the PC sequences of size $(N \times M)$.

III. Radar Processing Algorithm

A. Received Signals

1) Continuous Model: Considering a point target at distance R and the relative velocity V. We assume that $V > 0$ means the target is fleeing from the radar, and satisfies $2V/c \ll 1$, c is the speed of light. The received echo signal after down-conversion can be expressed as

$$y(t) = \sum_{n=0}^{N-1}\sum_{m=0}^{M-1} \rho w_n a_{nm} \text{rect}(t - mt_b - \tau) e^{-j2\pi f_c \tau} \times e^{j2\pi n\Delta f(t-\tau)} \quad (3)$$

where ρ is the scattering coefficient of the target, and

$$\tau = \frac{2(R - Vt)}{c} \quad (4)$$

TABLE I
DOPPLER PARAMETERS WITH DIFFERENT VELOCITIES AND BANDWIDTHS (f_c = 24GHz, t_b = 10us)

V (m/s)	f_d(KHz)	ξ	f_{dN}(KHz)	
			$B = 51.2$(MHz)	$B = 102.4$(MHz)
16	2.56	0.0256	0.0055	0.0109
50	8	0.08	0.0171	0.0341
100	16	0.16	0.0341	0.0683

2) Discrete Model: We assume that sampling frequency of D/A converter is $f_s = N/t_b$, the sampling time is $t = mt_b + it_b/N$, where $m = 0, \ldots, M-1$, $i = 0, \ldots, N-1$. The discrete form of $y(t)$ can be written as

$$y(i,m) = \sum_{n=0}^{N-1} \rho w_n a_{nm} e^{-j2\pi(f_c + n\Delta f)\frac{2(R - Vmt_b)}{c}} \times e^{j2\pi(n\Delta f + (n\Delta f + f_c)\frac{2V}{c})\frac{it_b}{N}} \quad (5)$$

Since $\frac{2V}{c} \ll 1$, $\frac{2V}{c} n\Delta f \ll n\Delta f$, equ(5) can be simplified as

$$y(i,m) = \sum_{n=0}^{N-1} \rho w_n a_{nm} e^{-j2\pi(f_c + n\Delta f)\frac{2R}{c}} e^{j2\pi ni/N} \Theta \quad (6)$$

where

$$\Theta = e^{j2\pi f_d m t_b} e^{j2\pi f_{dn} m t_b} e^{j2\pi\xi i/N} \quad (7)$$

and

$$f_d = f_c 2V/c \quad (8)$$

$$f_{dn} = n\Delta f 2V/c \quad (9)$$

$$\xi = f_d/\Delta f \quad (10)$$

B. Simplifications

Table I shows that values of the three Doppler parameters (f_d, $f_{dN} = B2V/c$, ξ) with different velocities and bandwidths at ISM band ($f_c = 24$GHz) and $t_b = 10$us. We assume that $f_c \gg B$. The values presented in Table I show that $f_d \gg f_{dN}$ and $\xi \approx 0$, we can get $\Theta \approx e^{j2\pi f_d m t_b}$. Hence, equ(5) can be further simplified as

$$y(i,m) = \sum_{n=0}^{N-1} \rho w_n a_{nm} e^{j2\pi ni/N} e^{-j2\pi(f_c + n\Delta f)\frac{2R}{c}} e^{j2\pi f_d m t_b} \quad (11)$$

which can be expressed in terms of matrix operations

$$\mathbf{y} = \phi \mathbf{w} \mathbf{F^{-1}} \mathbf{\Gamma} \mathbf{a} \mathbf{D} \quad (12)$$

where $\mathbf{y}$ is the matrix of size $(N \times M)$ representing the received echo, and

$$\mathbf{w} = \text{diag}\{w_0\ w_1 \ldots w_{N-1}\} \quad (13)$$

$$\mathbf{F^{-1}} = \begin{bmatrix} 1 & 1 & \ldots & 1 \\ 1 & \beta & \ldots & \beta^{N-1} \\ 1 & \beta^2 & \ldots & \beta^{2(N-1)} \\ \vdots & \vdots & & \vdots \\ 1 & \beta^{2(N-1)} & \ldots & \beta^{(N-1)^2} \end{bmatrix} \quad (14)$$

Authorized licensed use limited to: N.C. State University Libraries - Acquisitions & Discovery S. Downloaded on February 18,2026 at 03:32:02 UTC from IEEE Xplore. Restrictions apply.

$$\mathbf{\Gamma} = \text{diag}\{1 \ \gamma \ \gamma^2 \dots \gamma^{N-1}\} \tag{15}$$

$$\mathbf{a} = \begin{bmatrix} a_{00} & a_{01} & \dots & a_{0(M-1)} \\ a_{10} & a_{11} & \dots & a_{1(M-1)} \\ \vdots & \vdots & & \vdots \\ a_{(N-1)0} & a_{(N-1)1} & \dots & a_{(N-1)(M-1)} \end{bmatrix} \tag{16}$$

$$\mathbf{D} = \text{diag}\{1 \ \eta \ \eta^2 \dots \eta^{M-1}\} \tag{17}$$

and

$$\phi = \rho e^{-j2\pi f_c \frac{2R}{c}} \ \beta = e^{j2\pi/N} \ \gamma = e^{-j2\pi\Delta f \frac{2R}{c}} \ \eta = e^{j2\pi f_d t_b} \tag{18}$$

In equ(14), $\mathbf{F^{-1}}$ is the inverse discrete Fourier transform (IDFT) matrix.

C. Range and Velocity Processing

With the expression obtained in equ(12), the processing to retrieve both the range and velocity information is illustrated as follows:

Step 1: Add DFT matrix $\mathbf{F}$ to the received echo expression in equ(12), we get

$$\mathbf{Fy} = \phi \mathbf{w\Gamma aD} \tag{19}$$

Step 2: With the prior knowledge of the transmitted PC sequences, the element-wise division is performed for $\mathbf{a}$ and the matrix resulting from the previous step, which can be expressed as

$$\mathbf{Z} = \frac{\mathbf{Fy}}{\mathbf{a}} = \phi \mathbf{w\Gamma D} \tag{20}$$

where

$$\mathbf{Z}_{pm} = \rho w_p e^{-j2\pi f_c \frac{2R}{c}} e^{-j2\pi p\Delta f \frac{2R}{c}} e^{j2\pi f_d m t_b} \tag{21}$$

Step 3: N-point IDFT for every column of $\mathbf{Z}$ is computed to obtain the range profile.

Step 4: M-point DFT is calculated for every row of the matrix resulting from the previous step. The resulting matrix represents a two-dimensional range-velocity image.

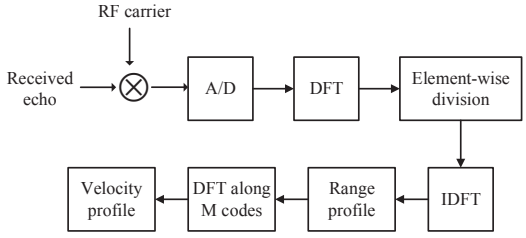


Fig. 1: Complete processing chain

The radar processing scheme is depicted in Fig. 1, it can be implemented by means of DFT and IDFT operations, hence, it is well-suited for practical applications. Moreover, due to the fact that a single pulse is transmitted, there is not exist the classical ambiguous range problem.

IV. Performance Analysis of the Proposed Algorithm

In this part, the performance of the radar processing algorithm is discussed. The performance analysis mainly includes the Doppler effect on the pulse compression loss and range estimation.

A. Pulse Compression Loss

Similar to analysis in [6], where the Doppler effect on such a scheme manifests itself as a loss in the pulse compression gain. The processing gain is $G_p = NM$, resulting from *Step 3* and *Step 4* in the proposed algorithm. There is no gain loss in the two *Steps*, this is because the range and velocity processing are completely independent. Hence, the gain loss mainly results from *Step 1* in the proposed algorithm. Hence, taking DFT of equ(6) concerning i when $V = 0$, we have

$$\begin{aligned} Y(p,m) &= N\rho w_p a_{pm} e^{-j2\pi(f_c + p\Delta f)\frac{2R}{c}} \\ & p = 0, \dots, N-1 \end{aligned} \tag{22}$$

Taking DFT of equ(6) concerning i when $V \neq 0$, we get

$$Y^{'}(p,m) = D(p,m) + \sigma(p,m) \quad p = 0, \dots, N-1 \tag{23}$$

where

$$\begin{aligned} D(p,m) =& \rho w_p a_{pm} e^{-j2\pi(f_c + p\Delta f)\frac{2(R - Vmt_b)}{c}} \\ & \times e^{j\pi\xi(N-1)/N} \frac{\sin(\pi\xi)}{\sin(\pi\xi/N)} \end{aligned} \tag{24}$$

$$\begin{aligned} \sigma(p,m) = & \sum_{\substack{n=0 \\ n\neq p}}^{N-1} \rho w_n a_{nm} e^{-j2\pi(f_c + n\Delta f)\frac{2(R - Vmt_b)}{c}} \\ & \times e^{j\pi(n-p+\xi)(N-1)/N} \frac{\sin(\pi(n-p+\xi))}{\sin(\pi(n-p+\xi)/N)} \end{aligned} \tag{25}$$

From equ(22)-(25), we get the expression for the compression loss $L_{pc}(n,p,\xi)$ due to Doppler as

$$L_{pc}(n,p,\xi) = \left| \frac{N \sin(\pi(n-p+\xi))}{\sin(\pi(n-p+\xi)/N)} \right| \tag{26}$$

which is a function of the subcarrier and ξ.

We assume $N = 1024$, Fig. 2 shows that $1/L_{pc}(n,p,\xi)$ varies with n and p when $\xi = 0.1$. It can be seen that $1/L_{pc}(n,p,\xi)$ obtains the maximum value when $n = p$. Also it demonstrates that the value of $|\sigma(p,m)|$ has little contribution to $|Y(p,m)|$. Hence, we can get

$$Y(p,m) = D(p,m) \tag{27}$$

The compression loss for various ξ when $n = p$ is plotted in Fig. 3. It can be seen that the compression loss increases as ξ rises. If we want to set a maximum allowable compression loss of 1dB, there is a limit to the target velocity. The mark shows that when $\xi = 0.261$, then the maximum Doppler frequency should satisfy $f_d \approx \Delta f/4$ when the maximum compression loss is less than 1dB. Therefore, the maximum target velocity can be calculated as $V_{\max} = \frac{c\Delta f}{8 f_c}$.

Authorized licensed use limited to: N.C. State University Libraries - Acquisitions & Discovery S. Downloaded on February 18,2026 at 03:32:02 UTC from IEEE Xplore. Restrictions apply.

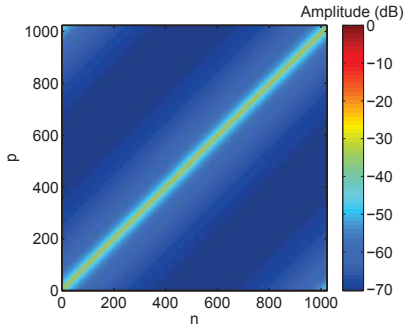


Fig. 2: $1/L_{pc}(n,p,\xi)$ versus n and p when $\xi = 0.1$

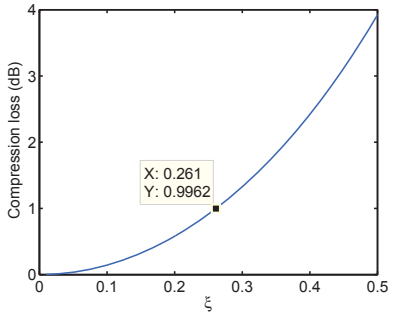


Fig. 3: The compression loss versus ξ when $n = p$

B. Doppler Effect on Range Estimation

Applying *Step 2* in the proposed algorithm, equ(27) can be changed to

$$\begin{aligned} Z(p,m) =& \phi w_p e^{j\pi\xi(N-1)/N} \frac{\sin(\pi\xi)}{\sin(\pi\xi/N)} \\ & \times k_R(p) \otimes k_V(m) \\ & p = 0,\ldots,N-1, m = 0,\ldots,M-1 \end{aligned} \tag{28}$$

where $\otimes$ represents dyadic product, and

$$k_R(p) = e^{-j2\pi p\Delta f\tau} \tag{29}$$

$$k_V(m) = e^{j2\pi m t_b f_{dp}} \tag{30}$$

Taking the IDFT of $k_R(p)$ concerning p, we get

$$r(q) = \frac{1}{N} e^{-j\pi(N-1)\Phi} \frac{\sin(N\pi\Phi)}{\sin(\pi\Phi)} \quad q = 0,\ldots,N-1 \tag{31}$$

where $\Phi = \Delta f \frac{2R}{c} - \frac{2V}{c} m - \frac{q}{N}$, which shows that the code number and target velocity introduce the error of range estimation. If $R \gg V m t_b$, we can get

$$R = \frac{cq}{2N\Delta f} \tag{32}$$

and the corresponding range resolution is $\Delta R = \frac{c}{2B}$.

V. Simulations and Analysis

Simulations are carried out to demonstrate the performance of the proposed processing algorithm using PC-OFDM integrated signal. The details of the system parameters are summarized in Table II. We use randomly generated communication data of size $(N \times \log_2 M)$ and m-sequence of length M to construct PC sequences. We assume $\rho = 1$ and the weight of subcarriers adopts the Hamming window function, adding in additive white Gaussian noise (AWGN).

TABLE II
PC-OFDM SYSTEM PARAMETERS

Symbol	Parameter	Value
f_c	RF carrier frequency	24GHz
N	Number of subcarriers	1024
M	Number of codes	256
t_b	Code duration	10us
Δf	Subcarrier separation	0.1MHz
B	Total signal bandwidth	102.4MHz
ΔR	Range resolution	1.4648m
ΔV	Velocity resolution	2.4414m/s
$V_{\max}$	Unambiguous velocity	$\pm$312m/s
SNR	Signal-to-noise ratio	10dB

A. Range Imaging

1) Performance Comparison: In order to show the improvement that is achieved by the proposed algorithm comparing the classic correlation processing [8]. Fig. 4 shows the normalized range profile of a point target with $R = 40\text{m}$ and $V = 0\text{m/s}$ obtained by the both algorithms. It can be seen that the proposed algorithm obtains better sidelobe characteristics, where the measured sidelobes is at a level of around -30dB with the cost of increased the mainlobe width. While for the correlation processing is at a level of around -12dB.

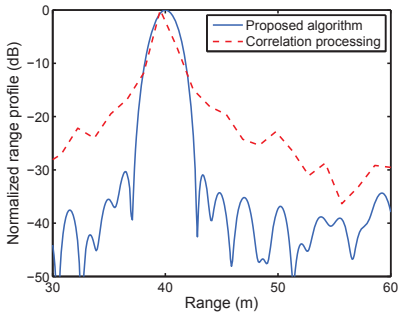


Fig. 4: Normalized range profile of a target with $R = 40\text{m}$ and $V = 0\text{m/s}$ based on the proposed algorithm and correlation processing

The complexities of the both algorithms are compared by using the number of complex multiplications, the former is $O(MN\log_2 N)$, while the latter is $O(MN\log_2(MN))$. Hence, the proposed algorithm has relative a low computational complexity.

Authorized licensed use limited to: N.C. State University Libraries - Acquisitions & Discovery S. Downloaded on February 18,2026 at 03:32:02 UTC from IEEE Xplore. Restrictions apply.

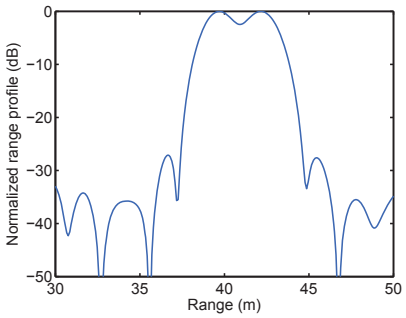


Fig. 5: Normalized range profile of two targets with $R = [40\ 41.9]\mathrm{m}$ and $V = [0\ 0]\mathrm{m/s}$

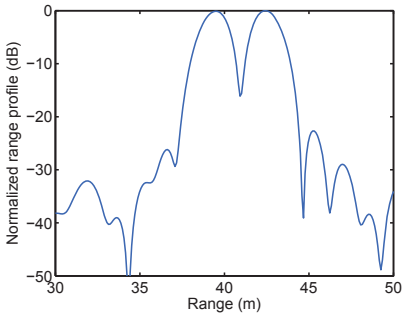
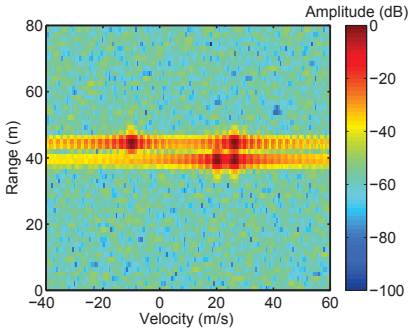
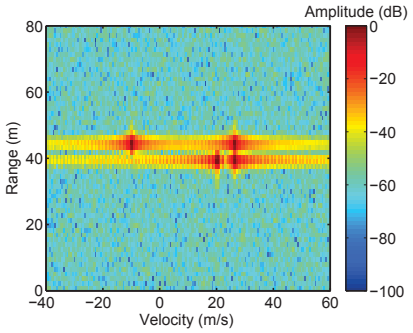


Fig. 6: Normalized range profile of two targets with $R = [40\ 41.9]\mathrm{m}$ and $V = [100\ 0]\mathrm{m/s}$



(a) $M = 256$



(b) $M = 512$

Fig. 7: Normalized radar profile of four targets with $R = [40\ 40\ 45\ 45]\mathrm{m}$ and $V = [20\ 26\ 26\ -10]\mathrm{m/s}$ at different code number

2) Doppler Effect on Range Estimation: Firstly, in order to evaluate the effectiveness of the proposed algorithm concerning the achievable resolution on the separability of adjacent targets. A simulation of two targets with $R = [40\ 41.9]\mathrm{m}$ and $V = [0\ 0]\mathrm{m/s}$ is performed, the corresponding normalized range profile is shown in Fig. 5. It can be seen that, the drop in the range profile between the two targets amounts approximately $3\mathrm{dB}$. Hence, under ideal conditions two targets in a distance of $\Delta d = 1.9\mathrm{m}$ are separable. That is because two targets must be a distance minimum of $\Delta R = 1.4648\mathrm{m}$ apart to be successfully resolved in this simulation. Also it is shown that the proposed algorithm is able to detect multiple adjacent targets.

Secondly, the effect of Doppler shift on the range estimation has been investigated. A similar simulation of the two targets with $R = [40\ 41.9]\mathrm{m}$ and $V = [100\ 0]\mathrm{m/s}$ is performed, the corresponding normalized range profile is shown in Fig. 6. The two targets are better separable, only the dynamic range has slightly decreased. Hence, range estimation is under little effect of the Doppler shift. This is an advantage for coded-OFDM radar signal, and has already been shown in [6].

B. Radar Imaging

In order to show that the proposed algorithm is able to estimate both range and velocity, independently, for multiple targets. We consider four targets with $R = [40\ 40\ 45\ 45]\mathrm{m}$ and $V = [20\ 26\ 26\ -10]\mathrm{m/s}$, and the corresponding range-velocity image at different code number is shown in Fig. 7.

As shown in Fig. 7, we can get that multiple targets can be separated even if they are located in the same distance or have the same velocity. The result shows that the estimation of range and velocity is completely independent and orthogonal. Besides, it is apparent that the velocity resolution improves as the code number increases. This is because the velocity resolution is $\Delta V = \frac{c}{2f_c M t_b}$. In order to achieve better velocity resolution, the code number should be large.

VI. Conclusion

In this paper, an integrated radar-communication system using PC-OFDM signals is presented. The designed waveform uses the transmitted information to control the cyclic-shift value of a phase sequence, that results in a relative high data rate. The processing algorithm within a single pulse is applied for range and velocity estimation, which eliminates the range ambiguity problem. Based on the theoretical analysis and simulation results, we can get (i) an expression for the compression loss is constructed due to the Doppler shift. Setting a limit to the allowed target velocity is essential in order to control the compression loss using one single pulse. (ii) The processing algorithm can achieve a high resolution range estimation and low computational complexity comparing the correlation processing. (iii) The range and velocity estimations are completely independent and the range-velocity coupling effects are eliminated. These conclusions

Authorized licensed use limited to: N.C. State University Libraries - Acquisitions & Discovery S. Downloaded on February 18,2026 at 03:32:02 UTC from IEEE Xplore. Restrictions apply.

and presented methods can provide insights to the design and operation of the integrated radar-communication systems.

ACKNOWLEDGMENT

This paper was supported by the National Natural Science Foundation of China (NSFC) under Grant (No. 61771159 and 91638204) and Guangdong Natural Science Foundation under Grant 2017A030313392.

REFERENCES

[1] S. D. Blunt, P. Yatham, and J. Stiles, "Intrapulse radar-embedded communications," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 46, no. 3, pp. 1185–1200, Jul. 2010.

[2] C. Sturm and W. Wiesbeck, "Waveform design and signal processing aspects for fusion of wireless communications and radar sensing," *P. IEEE.*, vol. 99, no. 7, pp. 1236–1259, Jul. 2011.

[3] C. Sturm, T. Zwick, and W. Wiesbeck, "An OFDM system concept for joint radar and communications operations," in *Proc. IEEE VTC'09*, Barcelona, Spain, Apr. 26–29, 2009, pp. 1–5.

[4] N. Levanon, "Multifrequency complementary phase-coded radar signal," *IEE P-RADAR SON NAV.*, vol. 147, no. 6, pp. 276–284, Dec. 2000.

[5] Y. Paichard, J. C. Castelli, P. Dreuillet, and G. Bobillot, "HYCAM: a RCS measurement and analysis system for time-varying targets," in *Proc. IEEE IMTC'06*, Sorrento, Italy, Apr. 24–27, 2006, pp. 921–925.

[6] G. Franken, H. Nikookar, and P. Van Genderen, "Doppler tolerance of OFDM-coded radar signals," in *Proc. EuRAD '06*, Manchester, UK, Sep. 13–15, 2006, pp. 108–111.

[7] K. Huo and J. Zhao, "The development and prospect of the new OFDM radar," *J Electr Inf Technol*, vol. 37, no. 11, pp. 2776–2789, Nov. 2015.

[8] D. Garmatyuk, J. Schuerger, Y. T. Morton, K. Binns, M. Durbin, and J. Kimani, "Feasibility study of a multi-carrier dual-use imaging radar and communication system," in *Proc. EuRAD '07*, Munich, Germany, Oct. 10–12, 2007, pp. 194–197.

[9] M. Ruggiano, E. Stolp, and P. V. Genderen, "Multi-target performance of LMMSE filtering in radar," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 48, pp. 170–179, Jan. 2012.

[10] J. F. Gu, J. Moghaddasi, and K. Wu, "Delay and Doppler shift estimation for OFDM-based radar-radio (RadCom) system," in *Proc. IEEE IWS'15*, Shanghai, China, Mar. 30–Apr. 1, 2015, pp. 1–4.

[11] A. Turlapaty, Y. Jin, and Y. Xu, "Range and velocity estimation of radar targets by weighted OFDM modulation," in *Proc. RadarConf '14*, Cincinnati, Ohio, USA, May 19–23, 2014, pp. 1358–1362.

[12] J. Lim, S. R. Kim, and D. J. Shin, "Two-step Doppler estimation based on intercarrier interference mitigation for OFDM radar," *IEEE ANTENN WIREL PR.*, vol. 14, pp. 1726–1729, Apr. 2015.

[13] Y. Liu, G. Liao, Z. Yang, and J. Xu, "A super-resolution design method for integration of OFDM radar and communication," *J. ELECTR INF TECHNOL*, vol. 38, no. 2, pp. 425–433, Feb. 2016.

[14] K. Chen, Y. Liu, and W. Zhang, "Study on integrated radar-communication signal of OFDM-LFM based on FRFT," in *Proc. IET Radar '15*, Hangzhou, China, Oct. 14–16, 2015, pp. 1–6.

[15] H. Lou, Y. Wu, Z. Ma, X. Zhao, and Q. Zhang, "A novel signal model for integration of radar and communication," in *Proc. IEEE ICCEM'16*, Guangzhou, China, Feb. 23–25, 2016, pp. 14–16.

[16] R. F. Tigrek, W. J. De Heij, and P. Van Genderen, "OFDM signals as the radar waveform to solve Doppler ambiguity," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 48, no. 1, pp. 130–143, Jan. 2012.

Authorized licensed use limited to: N.C. State University Libraries - Acquisitions & Discovery S. Downloaded on February 18,2026 at 03:32:02 UTC from IEEE Xplore. Restrictions apply.